Jenna M. Kline

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RESEARCH SUMMARY

Cyber-Physical Systems for Adaptive Ecological Sensing at the Far Edge

I build **cyber-physical systems** for autonomous environmental sensing in dynamic, resource-constrained field settings. My research advances **embodied intelligence**, **edge AI**, and **adaptive autonomy** to coordinate heterogeneous sensors and robots, enabling **cross-modal systems** to adapt what, where, and when they observe. I deploy these systems across **ecology**, **biodiversity monitoring**, and **coastal and climate resilience**, developing the foundations for persistent, adaptive observation of complex environmental systems.

EDUCATION AND TRAINING

Postdoctoral Associate, **Massachusetts Institute of Technology** 2026 — present
Department of Civil and Environmental Engineering; Climate Project at MIT
Advisor: Dr. Heidi Nepf

Ph.D. Computer Science & Engineering, **The Ohio State University** 2021 — 2026
Dissertation: Towards AI-Driven Multi-Modal Ecosystem Monitoring at the Far Edge
Major: Software Systems | ***Minors:*** Artificial Intelligence & Imageomics
Advisors: Dr. Christopher Stewart and Dr. Tanya Berger-Wolf

B.S. Industrial and Systems Engineering, *cum laude*, **The Ohio State University** 2014 — 2018
Honors in Integrated Business and Engineering (IBE)

FEATURED PUBLICATIONS

F featured, *J journal*, *C conference*, *P presentation*, *A artifact*; ** student mentee.* Note: Computer science conference publications are archival full papers that undergo rigorous peer review, as is typical for journal publications. Venue acceptance rate (**AR**) and impact factor (**JIF**) reported where available.

[F1] **FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets**
J. Kline, K. Meier, V. Shukla, E. G. A. Rolland, E. Iannino, et al.
Manuscript in preparation, **2026**. *Journal paper.*

[F2] **A Blueprint for Cross-Modal Coordination at the Far Edge**
J. Kline, A. Potlapally*, H. Subramoni, T. Berger-Wolf, and C. Stewart
ACM/IEEE Symposium on Edge Computing (SEC), **2026**. *Conference full paper (to appear).*

[F3] **A Maneuver-Indexed Testbed for Context-Aware Adaptive Wildlife Drones; J. Kline, *sole author***
IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C), **2026**.
Artifact paper. **ACSOS 2026 Best Artifact Award.**

[F4] **Studying Collective Animal Behaviour with Drones and Computer Vision**
J. Kline, S. Afridi, E. G. A. Rolland, G. Maalouf, L. Laporte-Devylder, et al.
Methods in Ecology and Evolution 16(10), **2025**. *Journal paper.* AR **25%**, JIF **6.2**.
Robert May Prize for best paper from early-career author.

[F5] **Edge-Native, Behavior-Adaptive Drone System for Wildlife Monitoring**
J. Kline, R. Katole, T. Berger-Wolf, and C. Stewart
DroneSys: Workshop on Autonomous Drone Computing Systems and Application at the ACM/IEEE Symposium on Edge Computing (SEC), **2025**. *Conference workshop paper.*
Associated work awarded **First Place in the Student Research Competition (SRC).**

[F6] **WildWing: An Open-Source, Autonomous and Affordable UAS for Animal Behaviour Video Monitoring**
J. Kline, A. Zhong*, K. Irizarry, C. V. Stewart, C. Stewart, et al.
Methods in Ecology and Evolution 17(2), **2026**. *Journal paper.* AR **25%**, JIF **6.2**.

[F7] **Characterizing and Modeling AI-Driven Animal Ecology Studies at the Edge**
J. Kline, A. O'Quinn, T. Berger-Wolf, and C. Stewart
ACM/IEEE Symposium on Edge Computing (SEC), **2024**. *Conference full paper.* **27%** AR.

[F8] **A Framework for Autonomic Computing for In Situ Imageomics**
J. Kline, C. Stewart, T. Berger-Wolf, M. Ramirez, S. Stevens, et al.
IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS), **2023**.
Conference vision paper. **23%** AR.

HONORS AND AWARDS

Best Artifact Award (2026): IEEE Autonomic Computing & Self-Organizing Systems (ACSOS) [F3]

Path of Professorship (2026): Competitive faculty-preparation program, MIT Career Advising & Professional Development

Robert May Prize (2025): Best early-career-author paper, British Ecological Society [F4] ($650)

Student Research Competition - First Place (2025): ACM/IEEE Symposium on Edge Computing [P12]

Departmental Research Award (2025): Outstanding Graduate Research in CSE, OSU

Hayes Forum - First Place (2025): Engineering Oral Presentation, Hayes Advanced Research Forum, OSU [P16] ($600)

Rising Stars in EECS (2025): MIT & Boston University Rising Stars Workshop

WiscProf Workshop (2025): Future Faculty in Engineering, Univ. of Wisconsin–Madison

Best Poster Award (2024): IEEE Autonomic Computing & Self-Organizing Systems (ACSOS) [C14]

TDAI Best Poster Award (2024): OSU Translational Data Analytics Institute Fall Forum [P20] ($300)

GRANTS AND FELLOWSHIPS

$51,800 in fellowship, research, and travel funding as a graduate student, plus a 20,000 GPU-hour NSF NAIRR compute allocation.

Presidential Fellowship (2025 — 2026): The Ohio State University Presidential Fellowship	$40,000
NSF NAIRR Pilot Allocation (2025): Digital Twins for Ecological Research (NAIRR250091)	20,000 GPU hours
Research Grant (2025): OSU Alumni Grant for Graduate Research and Scholarship	$3,200
Grace Hopper Celebration Scholarship (2022, 2023): OSU ACM-W Chapter, graduate student scholarship	$4,000
CRA-WP Grad Cohort for Women (2023): Computing Research Association travel award [P26]	$2,000
Student Travel Grant (2022, 2024): ACM/IEEE Symposium on Edge Computing	$2,600

RESEARCH EXPERIENCE

Postdoctoral Associate, Department of Civil and Environmental Engineering, MIT · **Climate Project at MIT** Jun 2026 — present
Multi-Scale Sensing for Coastal Salt-Marsh Resilience

- *Role:* Led the 2026 **drone field campaign** at Waquoit Bay NERR (team lead, FAA Part 107 pilot); developing edge-native drone workflows for vegetation monitoring.
- *Mentoring:* **Supervising four undergraduate researchers**.
- *Outputs:* **Student-led** refereed poster at **SEC 2026** [P1]; manuscripts in preparation.

Project lead, Department of Computer Science and Engineering, OSU 2025 — 2026
SmartWilds: Cross-Modal Coordination for Biodiversity Monitoring

- *Role:* **Led study design** and the 2025 **field campaign** at The Wilds with an undergraduate–PhD team; developed a **record–characterize–synthesize methodology** for provisioning cross-modal systems.
- *Funding:* **20,000 GPU-hour NSF NAIRR Pilot allocation**; **OSU Presidential Fellowship**.
- *Outputs:* **SEC 2026** full paper [F2]; **student-led** CV4Animals @ CVPR 2026 paper [C4]; Imageomics @ NeurIPS 2025 dataset paper [C7]; SmartWilds datasets and models [A7]; four TDWG 2026 talks.

Doctoral Researcher (dissertation research), OSU · **Visiting Doctoral Researcher, WildDroneEU consortium** 2023 — 2026
Autonomous, Behavior-Adaptive Drones for Animal Ecology

- *Role:* Built **WildWing**, an open-source autonomous UAS, and a **behavior-adaptive navigation pipeline**; released the **KABR** and **MMLA** collections (**360K+ downloads**).
- *Fieldwork:* Led **drone field campaigns** in Kenya (Mpala 2023; Ol Pejeta 2025, 2026) and Ohio (The Wilds 2025); WildDroneEU research visits (2024).
- *Funding & mentoring:* **OSU Alumni Grant for Graduate Research and Scholarship**; **co-supervised a master's thesis** (University of Bologna).
- *Outputs:* **Methods in Ecology and Evolution** 2026 [F6] and 2025 (**Robert May Prize**) [F4]; ACSOS 2026 **Best Artifact Award** [F3]; ACSOS 2024 **Best Poster Award** [C14]; SEC 2025 **Student Research Competition, first place** [P12]; DroneSys @ SEC 2025 [F5]; WACVW 2024 [C12]; CV4Animals @ CVPR 2026 [C5] and 2025 [C9]; WildWing [A4], KABR [A5], MMLA [A6].

Doctoral Researcher (dissertation research), OSU 2022 — 2026

Edge Infrastructure and AI-Ready Data for Ecological Sensing

- *Role:* Characterized **AI-driven animal ecology (ADAE) workloads**; developed **trace-replay and performance models** for edge configurations; led the **FAIR^2 Drones** data standard and **kabr-tools** pipeline.
- *Senior author:* Environment-aware dynamic pruning for distributed inference (**1.5× throughput, 3× SLO attainment**).
- *Outputs:* **SEC 2024** full paper [F7]; **IEEE EDGE 2025** full paper [C6]; ACSOS 2023 vision paper [F8]; Sim2Real2Field @ CoRL 2026 [C1]; FAIR^2 Drones manuscript in preparation [F1]; TDWG [P2], FARR [P10], and **invited ESA 2026** talks; FAIR^2 Drones [A1] and kabr-tools [A2] software.

Supercomputer Institute Graduate Intern, Los Alamos National Laboratory Summer 2023

High-Performance Computing Filesystem Characterization

- *Role:* Built, configured, and tested an HPC cluster from scratch; led the filesystem characterization project.
- *Outputs:* **SC23** Graduate Posters, **ACM Student Research Competition** [P22].

Dissertation research supported by the NSF Imageomics (2118240) and ICICLE (2112606) Institutes, the NSF–NSERC AI and Biodiversity Change (ABC) Center (2330423), an OSU Presidential Fellowship, and an OSU Alumni Grant for Graduate Research.

Remaining research outputs (journals, conferences, presentations, artifacts) included in the research outputs section (p. 4).

MENTORING AND TEACHING

Mentoring and Student Supervision

Mentoring focus: field research (deployment planning, drone operations, field safety), software engineering (Git/GitHub workflows, code review, documentation, reproducible pipelines), scientific writing (drafting and revising papers), and scientific communication (conference posters and talks).

Massachusetts Institute of Technology (2026–present). Supervised four undergraduate researchers through the Undergraduate Research Opportunities Program (UROP) on a salt-marsh monitoring study using drones and computer vision, including drone fieldwork at the Waquoit Bay National Estuarine Research Reserve. One student-led project accepted to SEC 2026 poster session (42% AR) [P1].

The Ohio State University (2022–2026). Supervised five undergraduate researchers, including a team conducting fieldwork at The Wilds Conservation Center, and led a field-campaign team of undergraduate, master's, and PhD students. Student-led projects were published at CV4Animals at CVPR 2026 [C4, C5] and SEC 2025 [P13].

University of Bologna (2024). Co-supervised Denys Grushchak's master's thesis, *Herd Monitoring with Autonomous Drones: A Decentralized k-Coverage-Inspired Approach*, with Dr. Danilo Pianini, advising on ecological considerations and experimental design; the thesis resulted in two publications at the IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS) [C14, C15].

Teaching Experience

*All courses taught at The Ohio State University, Department of Computer Science and Engineering. Student evaluations (CSE 2111): overall teaching effectiveness **4.5/5.0 and 4.4/5.0**, above department, college, and university averages.*

Lab Instructor — *CSE 2111: Modeling and Problem Solving with Spreadsheets and Databases*. Undergraduate required course. Led two 35-student lab sections twice weekly and supervised a team of undergraduate graders. Topics: spreadsheet and database modeling, programming concepts, business applications.

Graduate Teaching Assistant — *CSE 3244: Data Management in the Cloud*. Senior-level elective. Sole graduate teaching assistant responsible for grading and supervising lab assignments and the final project. Topics: data organization on cloud computing architectures, B-tree and hash-based indexing, query optimization fundamentals, data partitioning, and distributed task scheduling.

SERVICE

Program Committee Member: AI4Nature Workshop on Computer Vision and AI for Environmental Monitoring, Biodiversity Analysis, and Ecosystem Restoration, IEEE AVSS 2026

Organizing Committee Member: Community Building Co-Chair, ACSOS 2026, Artifact Co-Chair, ACSOS 2027 2026 — present

Workshop Organizer & Lead Facilitator: Edge AI for Conservation workshop, International Conservation on Technology for Conference (ICTC), Lima, Peru 2026

Co-founder and Co-lead: Edge Computing for Conservation Group, WILDLABS Network 2025 — present

Workshop Organizer: Imageomics Image Datapalooza Hackathon 2023

Volunteer: Code I/O coding camp, The Ohio State University ACM-W Chapter 2023

Peer Mentor: Graduate Student Peer Mentoring Program, The Ohio State University 2022 — 2024

Journal Reviewer: International Journal of Computer Vision; Methods in Ecology and Evolution; Scientific Reports; Ecological Infor-

matics; Ecological Modelling; Global Ecology and Conservation; Marine Mammal Science; PLOS ONE; IET Computer Vision; Ecology and Evolution; MethodsX.

SKILLS

Field Research and Leadership: FAA Part 107 Remote Pilot Certificate (small UAS) · drone piloting and flight operations · field campaign planning, training, and coordination (Kenya, Ohio, Massachusetts)

Autonomous Sensing: autonomous flight control · mission planning · perception–control integration · adaptive navigation policies · multimodal sensor networks (drone, camera trap, ARU)

Edge AI and Distributed Computing: real-time AI inference on resource-constrained hardware · edge and embedded platforms · hardware-aware system design · edge–cloud orchestration · workload and performance characterization · shared HPC environments · distributed storage and filesystems · wildlife detection, tracking, pose estimation, behavior recognition, and re-identification with AI, including edge inference on drones and embedded devices

Data Infrastructure: FAIR and AI-ready metadata standards · multimodal dataset design and curation · spatiotemporal data synchronization · annotation workflows (GitHub) · reproducible data releases and benchmarking (Hugging Face, Zenodo)

Programming and Tools: Python · PyTorch · SQL · Linux · Git · scientific computing and data-analysis workflows

RESEARCH OUTPUTS

Continuation of featured publications (p. 1). Given the interdisciplinary and collaborative nature of my work, this section lists computer science conference papers, journal publications, and associated preprints. Computer science conference publications (SEC, EDGE, etc.) are archival full papers that undergo rigorous peer review, as is typical for journal publications. Accepted ecology conference abstracts are listed under Presentations. Venue acceptance rate (**AR**) and impact factor (**JIF**) reported where available.

Journal Articles

[J1] **When to Trust Automation for In Situ Video Animal Behavior Monitoring**
J. Kline, B. Braasch, B. Fischer, A. Zhong*, M. Thompson, et al.
Manuscript under review, **2026**.

[J2] **Putting the "Adaptive" in Adaptive Monitoring: From Fast Data to Meaningful Ecological Change**
L. J. Pollock, P. H. P. Braga, C. R. Florian, K. Hébert, **J. Kline**, et al.
Manuscript in minor revision; EcoEvoRxiv preprint, **2026**.

[J3] **Simulated and Field-Based Error Characterization of Animal Geolocalisation and Relative Positioning via Commercial Drones**
K. Meier, E. G. A. Rolland, E. Iannino, M. Simpson, **J. Kline**, et al.
Manuscript under review, **2026**.

[J4] **Simulation-Grounded Decentralized Multi-UAV Coordination for Scalable In-Situ Imageomics of Wildlife Herds**
G. Aguzzi, M. Baiardi, N. Farabegoli, D. Grushchak*, **J. Kline**, D. Pianini, C. Stewart, and M. Viroli
Manuscript under review, **2026**.

[J5] **SORA 2.5-Guided BVLOS UAS for Wildlife Conservation in Kenya: Reducing Friction Between Safety and Field Operations**
G. Maalouf et al., including **J. Kline**
Drones 10(3), article 178, **2026**. JIF **5.2**.

[J6] **BaboonLand Dataset: Tracking Primates in the Wild and Automating Behaviour Recognition from Drone Videos**
I. Duporge, M. Kholiavchenko, et al., including **J. Kline**
International Journal of Computer Vision 133(9), **2025**. JIF **10.3**.

[J7] **Deep Dive into KABR: A Dataset for Understanding Ungulate Behavior from In-Situ Drone Video**
M. Kholiavchenko, **J. Kline**, M. Kukushkin, O. Brookes, S. Stevens, et al.
Multimedia Tools and Applications 84(21), **2025**.

[J8] **Impact of Drone Disturbances on Wildlife: A Review**
S. Afridi, L. Laporte-Devylder, G. Maalouf, **J. Kline**, S. G. Penny, et al.
Drones 9(4), article 311, **2025**. JIF **5.2**.

Conference and Workshop Articles

[C1] **Dreams Are Cheaper Than Flights: Curated World-Model Data Improves Aerial Detection**
R. Katole, V. Y. Patil, **J. Kline**, and C. Stewart
Sim2Real2Field Workshop, Conference on Robot Learning (CoRL), **2026**.
Workshop paper (to appear).

[C2] **What the Sheep Can Teach the Shepherd: A Vision for Self-Organizing Drone Swarms Informed by Collective Animal Behavior**
J. Kline, S. Stevens, D. I. Rubenstein, T. Berger-Wolf, and C. Stewart
IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS), **2026**.

Vision paper.

[C3] **WildBox: A Dataset and Benchmark for Aerial Monocular 3D Detection of African Savanna Wildlife**
V. Shukla, K. Meier, L. Laporte-Devylder, C. R. Saint-Jean, **J. Kline**, et al.
Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, **2026**.
Full dataset paper. AR **25.8%**.

[C4] **Cross-Modal Corroboration for Annotation-Free Wildlife Monitoring**
B. Pillai*, V. Viswapriyan*, C. Stewart, T. Berger-Wolf, and **J. Kline**
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2026**.
Workshop paper.

[C5] **Autonomous UAV Navigation for Individual Wildlife Re-Identification**
C. Sun*, T. Berger-Wolf, and **J. Kline**
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2026**.
Workshop paper.

[C6] **Environment-Aware Dynamic Pruning for Pipelined Edge Inference**
A. O'Quinn, C. Snedeker, S. Zhang, and **J. Kline**
IEEE International Conference on Edge Computing and Communications (EDGE), **2025**.
Full paper. **26%** AR.

[C7] **SmartWilds: Multimodal Wildlife Monitoring Dataset**
J. Kline, A. Potlapally*, B. Pillai*, T. Wani, R. Katole, et al.
Third Workshop on Imageomics at NeurIPS, **2025**.
Workshop paper.

[C8] **Mind the (Data) Gap: Evaluating Vision Systems in Small Data Applications**
S. Stevens, S. M. Rayeed, and **J. Kline**
Third Workshop on Imageomics at NeurIPS, **2025**.
Workshop paper.

[C9] **MMLA: Multi-Environment, Multi-Species, Low-Altitude Aerial Footage Dataset**
J. Kline, S. Stevens, G. Maalouf, C. R. Saint-Jean, D. N. Ngoc, et al.
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2025**.
Workshop paper.

[C10] **WildLive: Near Real-Time Visual Wildlife Tracking Onboard UAVs**
N. N. Dat, T. Richardson, M. Watson, K. Meier, **J. Kline**, et al.
CV4Animals Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), **2025**.
Workshop paper.

[C11] **An Adaptive Autonomous Aerial System for Dynamic Field Animal Ecology Studies**
J. Kline
IEEE ACSOS Companion (ACSOS-C), **2024**.
PhD Forum extended abstract, presented as talk.

[C12] **KABR: In-Situ Dataset for Kenyan Animal Behavior Recognition from Drone Videos**
M. Kholiavchenko, **J. Kline**, M. Ramirez, S. Stevens, A. Sheets, et al.
IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), **2024**.
Workshop paper.

[C13] **Integrating Biological Data into Autonomous Remote Sensing Systems for In Situ Imageomics: A Case Study for Kenyan Animal Behavior Sensing with Unmanned Aerial Vehicles**
J. Kline, M. Kholiavchenko, O. Brookes, T. Berger-Wolf, C. V. Stewart, et al.
First Workshop on Imageomics at AAAI, **2024**.
Workshop paper.

[C14] **Decentralized Multi-Drone Coordination for Wildlife Video Acquisition**
D. Grushchak*, **J. Kline**, D. Pianini, N. Farabegoli, G. Aguzzi, et al.
IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS), **2024**.
Full paper. **24%** AR. **ACSOS 2024 Best Poster Award.**

[C15] **An Agent-Based Model of Directional Multi-Herds**
D. Grushchak*, **J. Kline**, D. Pianini, and N. Farabegoli
IEEE ACSOS Companion (ACSOS-C), **2024**.
Workshop paper.

[C16] **Drone Swarms for Animal Monitoring: A Method for Collecting High-Quality Multi-Perspective Data**
E. Rolland, K. A. R. Grøntved, L. Laporte-Devylder, **J. Kline**, U. P. S. Lundquist, et al.
15th International Micro Air Vehicle Conference (IMAV), **2024**. *Full paper.*

PRESENTATIONS AND INVITED ENGAGEMENTS *Sole author unless otherwise noted.*

[P1] **Toward an Edge-Native Drone System for Coastal Salt Marsh Monitoring**
J. Xiang*, E. I. H. Lee*, H. Nepf, and **J. Kline**
Refereed extended abstract, presented as poster, ACM/IEEE Symposium on Edge Computing (SEC), **2026**.
Proceedings to appear. **42%** AR.

[P2] **FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P3] **A Metadata-First FAIR4AI Checklist and Automated Evaluation Agent for Biodiversity Datasets**
E. Sokol et al., including **J. Kline**
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P4] **Cross-Modal Coordination for Biodiversity Monitoring at Scale: Lessons from SmartWilds**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P5] **Digital Twins as System-Optimization Tools: Provisioning Edge and Cloud Infrastructure for Biodiversity Monitoring at Scale**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P6] **kabr-tools: A Modular Workflow for Automated Video-Based Behavioral Monitoring Across the Autonomy Spectrum**
J. Kline et al.
Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, **2026**.

[P7] **AI for Discovery: Ecosystem & Biodiversity Science**
Invited roundtable participant, Aspen Institute Science & Society Program, **2026**. Roundtable insights synthesized in the report *AI for Discovery: New Frontiers, Challenges, and Opportunities for Scientific Research*.

[P8] **Lessons from FAIR^2 Drones: Toward AI-Ready Multimodal, Multispatial Sensor Data for Ecology**
Invited symposium talk, "Building an AI-Ready Ecology and Biodiversity Data Infrastructure for Science and Action," Ecological Society of America (ESA) Annual Meeting, Salt Lake City, UT, **2026**.

[P9] **AI for Ecology: Accelerating Discoveries, Reducing Uncertainties, and Scaling Solutions**
J. Kline, panelist
Special Session SS23, Ecological Society of America (ESA) Annual Meeting, Salt Lake City, UT, **2026**.

[P10] **FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets**
J. Kline et al.
Poster presentation, FARR Workshop: FAIR in ML, AI Readiness and Reproducibility, Washington, DC, **2026**.

[P11] **Autonomous Drones for Animal Studies with Edge AI**
Invited talk, Global Conservation Tech & Drone Forum, Nairobi, Kenya, **2026**.

[P12] **An Edge-Native Approach to Behavior-Adaptive Navigation in Drone Systems**
J. Kline, R. Katole, and C. Stewart
Refereed extended abstract, presented as poster, ACM/IEEE Symposium on Edge Computing (SEC), **2025**.
SEC 2025 Student Research Competition — First Place.

[P13] **An Edge-to-Cloud Framework for Vigilance-Adaptive Drones**
P. Covey*, **J. Kline**, and C. Stewart
Refereed extended abstract, presented as poster, ACM/IEEE Symposium on Edge Computing (SEC), **2025**.

[P14] **Autonomous AI-Driven Ecological Sensing Systems**
Poster presentation, MIT and Boston University Rising Stars in EECS Workshop, **2025**.

[P15] **How Do Drones Fit into Multimodal Sensing Networks?**
Poster presentation, WildDrone Summer School, Max Planck Institute of Animal Behavior, Konstanz, Germany, **2025**.

[P16] **Autonomous, Adaptive Vision-Based Remote Sensing System for Dynamic Field Animal Ecology Studies**
Oral presentation, Edward F. Hayes Advanced Research Forum, The Ohio State University, **2025**.
First Place, Engineering Oral Presentation.

[P17] **Drones and AI in Field Animal Ecology: Integrating Biology, Animal Science, and Technology**
Invited lecture, Fridays at Findlay, University of Findlay, **2025**.

[P18] **From Ohio to Kenya: Autonomous Drones in Field Ecology and Conservation**
Invited lecture, Conservation Science Symposium, Muskingum University and The Wilds Conservation Center, **2025**.

[P19] **Autonomous Drones for Collective Wildlife Behavior**
Invited lecture, WildDrone Seminar Series, **2025**.

[P20] **Individual Identification of Zebras with Autonomous UAV Swarms**
Poster presentation, TDAI Interdisciplinary Research Fall Forum, The Ohio State University, **2024**.
Best Poster Award.

[P21] **Adaptive, Autonomous Aerial Systems for Dynamic Field Animal Ecology Studies**
Invited lecture, University of Bologna, Cesena, Italy, **2024**.

[P22] **Seeing the Trees for the Forest: Describing HPC Filesystem Trees with the Grand Unified File-Index (GUFI)**
J. Kline, J. Lee, and R. Davis
Refereed extended abstract, poster presentation, ACM Student Research Competition (SRC), International Conference for High Performance Computing, Networking, Storage, and Analysis (SC23), Denver, CO, **2023**.

[P23] **Intelligent Cyberinfrastructure for Remote Sensing with Autonomous Aerial Robotics**
Poster presentation, TDAI Interdisciplinary Research Fall Forum, The Ohio State University, **2023**.

[P24] **Interdisciplinary Applications of Autonomous Unmanned Aerial Vehicles**
Poster presentation, STARS Celebration at Tapia and Ohio Celebration of Women in Computing, **2023**.

[P25] **Autonomous UAV Missions for Studying Wildlife Behavior: A Case Study for the Individual Identification of Zebras**
Poster presentation, Midwest Machine Learning Symposium (MMLS), Chicago, IL, **2023**.

[P26] **Individual Identification of Zebras with Autonomous UAV Swarms**
Poster presentation, CRA-WP Grad Cohort for Women, San Francisco, CA, **2023**.

[P27] **Edge Computing for Software-Defined Cartography**
PhD Forum extended abstract, presented as talk, ACM/IEEE Symposium on Edge Computing (SEC), Seattle, WA, **2022**.

ARTIFACTS: MODELS, DATASETS, AND SOFTWARE

[A1] **FAIR^2 Drones: Data Infrastructure for AI-Ready Wildlife Drone Datasets**
Software and datasets: GitHub, Hugging Face. *FAIR, AI-ready, Darwin Core–compliant data infrastructure for multimodal animal ecology datasets. Described in [F1, P2, P10].*

[A2] **kabr-tools: Automated Framework for Multi-Species Behavioral Monitoring**
Software: GitHub. *Open-source pipeline for analyzing animal behavior from in situ video. Used in [J1, P6].*

[A3] **A Maneuver-Indexed Evaluation Framework for Context-Aware Adaptive Wildlife Drones**
Software and data: Hugging Face Spaces. *Interactive research demo and evaluation data. Companion to [F3].*
***ACSOS 2026 Best Artifact Award** (Grants and Awards, p. 2).*

[A4] **WildWing: Open-Source Autonomous UAS for Animal Behaviour Monitoring**
Software: GitHub. *Open-source drone system for autonomous animal monitoring. Companion to [F6].*

[A5] **In Situ Kenyan Animal Behavior Recognition (KABR) Datasets and Models Collection**
Datasets and models: Hugging Face. *Role:* data collection, annotation, analysis, telemetry integration, and public release. *Drone videos, annotations, synchronized telemetry, and models. Used in [J1, J4, J7, C12, C15].*
164K downloads

[A6] **Multi-Environment, Multi-Species, Low-Altitude Aerial (MMLA) Datasets and Models Collection**
Datasets and models: Hugging Face. *Role:* data collection, annotation, analysis, curation, and public release. *Multi-site detection and pose datasets, and detection and pose models trained on them. Used in [F3, C9].*
204K downloads

[A7] **SmartWilds: Multimodal Wildlife Monitoring Datasets and Models Collection**
Datasets and models: Hugging Face. *Role:* project lead; led study design, field data collection, annotation, analysis, multimodal coordination, and public release. *Camera-trap, bioacoustic, and drone datasets. Used in [F2, C4, C7].*
13K downloads

REFERENCES

Dr. Heidi Nepf — *Postdoctoral Supervisor*
Donald and Martha Harleman Professor of Civil and Environmental Engineering
Massachusetts Institute of Technology
hmnepf@mit.edu

Dr. Tanya Berger-Wolf — *PhD co-advisor*
Professor of Computer Science and Engineering
The Ohio State University
Director, Translational Data Analytics Institute
Director, NSF Imageomics Institute
berger-wolf.1@osu.edu

Dr. Christopher Stewart — *PhD co-advisor*
Professor of Computer Science and Engineering
The Ohio State University
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